Vector Spaces III: Sums, Intersections and Direct Sums

Problem sheet for the Day 4 class · CSIR NET / GATE / IIT JAM previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. GATE 2013 · NAT

Let $W_1 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = x_2 + 2x_3\}$ and $W_2 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : x_1 = 3x_2 + 2x_3\}$. Then $\dim(W_1 + W_2) =$ _____.

2. JAM 2019 · NAT

Let $W_1 = \{x \in \mathbb{R}^{100} : x_i = 0 \text{ whenever } 4 \mid i\}$ and $W_2 = \{x \in \mathbb{R}^{100} : x_i = 0 \text{ whenever } 5 \mid i\}$. Then $\dim(W_1 \cap W_2) =$ _____.

3. CSIR NET Dec 2018 · MCQ

Let $W_1 = \{(u, v, w, x) \in \mathbb{R}^4 : u + v + w = 0,\ 2v + x = 0,\ 2u + 2w - x = 0\}$ and $W_2 = \{(u, v, w, x) \in \mathbb{R}^4 : u + w + x = 0,\ u + w - 2x = 0,\ v - x = 0\}$. Then which among the following is true?

(A) $\dim(W_1) = 1$
(B) $\dim(W_2) = 2$
(C) $\dim(W_1 \cap W_2) = 1$
(D) $\dim(W_1 + W_2) = 3$

4. CSIR NET Dec 2018 · MCQ

Consider the subspaces $W_1 = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ and $W_2 = \{(x, y, z) \in \mathbb{R}^3 : x - y + z = 0\}$. If W is a subspace of $\mathbb{R}^3$ such that (i) $W \cap W_2 = \text{span}\{(0, 1, 1)\}$, (ii) $W \cap W_1$ is orthogonal to $W \cap W_2$ with respect to the usual inner product of $\mathbb{R}^3$, then

(A) $W = \text{span}\{(0, 1, -1), (0, 1, 1)\}$
(B) $W = \text{span}\{(1, 0, -1), (0, 1, -1)\}$
(C) $W = \text{span}\{(1, 0, -1), (0, 1, 1)\}$
(D) $W = \text{span}\{(1, 0, -1), (1, 0, 1)\}$

5. CSIR NET Dec 2016 · MSQ

Let W_1, W_2, W_3 be three distinct subspaces of $\mathbb{R}^{10}$ such that each W_i has dimension 9. Let $W = W_1 \cap W_2 \cap W_3$. Then we can conclude that

(A) W may not be a subspace of $\mathbb{R}^{10}$
(B) $\dim W \leq 8$
(C) $\dim W \geq 7$
(D) $\dim W \leq 3$

6. CSIR NET Dec 2014 · MSQ

For arbitrary subspaces U, V and W of a finite dimensional vector space, which of the following hold?

(A) $U \cap (V + W) \subset U \cap V + U \cap W$
(B) $U \cap (V + W) \supset U \cap V + U \cap W$
(C) $(U \cap V) + W \subset (U + W) \cap (V + W)$
(D) $(U \cap V) + W \supset (U + W) \cap (V + W)$

7. Exam pattern MSQ Let S, T be subspaces of $\mathbb{R}^{24}$ with $\dim S = 19$ and $\dim T = 17$. Which of the following are true?

(A) the smallest possible $\dim(S \cap T)$ is 2

(B) the largest possible $\dim(S \cap T)$ is 18

(C) the smallest possible $\dim(S + T)$ is 19

(D) the largest possible $\dim(S + T)$ is 24

8. Exam pattern MSQ In $M_n(\mathbb{R})$, $n \geq 2$, which pairs of subspaces give $M_n(\mathbb{R}) = W_1 \oplus W_2$?

(A) symmetric, skew-symmetric

(B) upper triangular, lower triangular

(C) upper triangular, strictly lower triangular

(D) trace-zero, scalar matrices

9. Exam pattern NAT Let $W_1 = \text{span}\{(1,0,1,0), (0,1,0,1)\}$ and $W_2 = \text{span}\{(1,1,0,0), (0,0,1,1)\}$ in $\mathbb{R}^4$. Then $\dim(W_1 \cap W_2) =$ _____.

10. Exam pattern MSQ Let $W = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$. Which of the following lines are complements of W in $\mathbb{R}^4$?

(A) $\text{span}\{(1,1,1,1)\}$

(B) $\text{span}\{(1,-1,0,0)\}$

(C) $\text{span}\{(1,0,0,0)\}$

(D) $\text{span}\{(1,1,-1,-1)\}$